

Raman maps of brain tissue separate HSP70-related stress states: a gradient boosting study with an audit of its own validation

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Abstract

Raman spectroscopy reads the molecular composition of tissue without labels or damage, which makes it attractive for studying how the brain responds to stress. We analysed confocal Raman maps of mouse brain tissue from three experimental groups: controls, animals with endogenous expression of the chaperone HSP70 under stress, and animals that received exogenous HSP70. Each map was cleaned with asymmetric least squares baseline removal, Savitzky–Golay smoothing and standard normal variate scaling, classified pixel by pixel with LightGBM, and turned into one decision per map by averaging pixel probabilities. On held-out maps the model reached 83% accuracy in the 930–2000 cm^{-1} window and 79% in the 2600–3200 cm^{-1} window, against a chance level of 33%. Feature attribution shows that both models rely mainly on lipid bands: even in the fingerprint window, almost 60% of the split gain falls on CH_2 and C–C skeletal vibrations of acyl chains, while protein-specific bands receive about 7%. This points to membrane remodelling as the main spectral signature separating the groups, in line with the known association of HSP70 with lipid membranes. We also audit our own evaluation. Maps were split by file rather than by animal, early stopping used the held-out maps, and the test set holds only 24 maps per window. A team that evaluated the same data with held-out animals reported far lower scores, so our numbers should be read as an upper bound. We describe what a leakage-free protocol for this dataset should look like.

1 Introduction

Raman spectroscopy measures inelastic light scattering from molecular vibrations. A single spectrum carries information about proteins, lipids, nucleic acids and other components of a sample, and confocal instruments can collect thousands of such spectra across a tissue section in one scan [1, 2]. In neuroscience and neuro-oncology this has made Raman mapping a practical tool for label-free tissue characterisation, starting with the mapping of native brain tissue and tumours [3].

The spectra themselves are hard to read by eye. Tissue autofluorescence adds a broad background that is often stronger than the Raman peaks, detector noise is significant, and the differences between biological conditions are subtle shifts in many overlapping bands. For this reason Raman analysis has become closely tied to chemometrics and machine learning [4]. Recent reviews stress that classification accuracy alone is not enough: models should be interpretable, biologically plausible and validated in a way that reflects how they will be used [5].

This study looks at the heat shock protein HSP70. HSP70 is a molecular chaperone that helps proteins fold, prevents aggregation and supports cell survival under stress [6]. Beyond its role inside the cell, HSP70 associates with lipid membranes and can help restore membrane stability during stress [7], and exogenous HSP70 delivered to mice has been shown to have neuroprotective effects [8]. The question we ask is whether brain tissue from animals with endogenous HSP70 expression, animals given exogenous HSP70, and controls can be told apart from Raman maps, and which spectral regions carry the difference.

The work was done during the Nuclear IT Hack 2026 hackathon. In this preprint we restate it in a more complete form. Our contributions are:

- a simple and reproducible pipeline from raw Raman maps to a per-sample decision, built from standard preprocessing steps and a gradient boosting classifier;
- a spectral interpretation of the trained models, which shows that lipid bands dominate the decision in both spectral windows;
- an explicit audit of our evaluation protocol, which identifies three sources of optimistic bias and outlines how to remove them.

2 Data

The dataset was provided by the hackathon organisers. Spectra were recorded on a Renishaw inVia Qontor confocal Raman microscope from sections of three brain regions (cortex, striatum and cerebellum). Each file is a map: a grid of pixels, each with its own spectrum. Every map belongs to one of three groups:

- **Control:** tissue from untreated animals;
- **Endo:** tissue under stress with endogenous expression of HSP70;
- **Exo:** tissue from animals that received exogenous HSP70.

Maps were acquired in two spectral windows that we modelled separately. The *fingerprint window* (930–2000 cm^{-1} , referred to as 1500) contains bands of proteins, lipids and nucleic acids. The *high wavenumber window* (2600–3200 cm^{-1} , referred to as 2900) is dominated by C–H stretching of lipids and proteins. In total we processed 241 maps, about 120 per window, which gave roughly 63 000 spectra per window with a near-balanced class distribution. The maps come from a small number of mice, on the order of ten [9], which turns out to matter for evaluation (Section 5.3).

The raw data are not public and are not included with this preprint.

3 Methods

The full pipeline is shown in Figure 1. All steps are applied to each pixel spectrum independently, so preprocessing cannot leak information between training and test maps.

3.1 Resampling

Maps differ slightly in their wavenumber axes. Each spectrum was linearly interpolated onto a fixed grid of 1000 points, 930–2000 cm^{-1} for the fingerprint window and 2600–3200 cm^{-1} for the high wavenumber window.

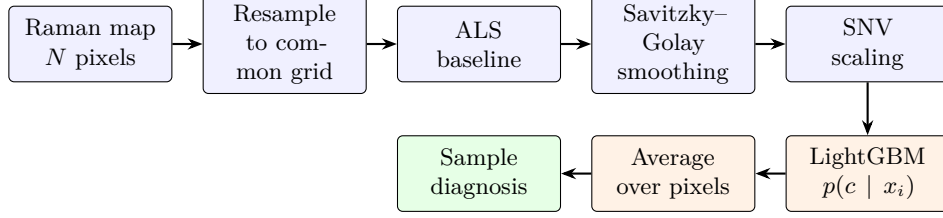


Figure 1: Processing pipeline. Every pixel spectrum is cleaned and classified independently; the map-level decision is the class with the highest mean probability.

3.2 Baseline removal

Autofluorescence produces a smooth, slowly varying background. We estimated it with asymmetric least squares (ALS) [10]. For a spectrum $y \in \mathbb{R}^L$ the baseline z minimises

$$S(z) = \sum_{i=1}^L w_i (y_i - z_i)^2 + \lambda \sum_{i=3}^L \left(\Delta^2 z_i \right)^2, \quad \Delta^2 z_i = z_i - 2z_{i-1} + z_{i-2}, \quad (1)$$

which leads to the sparse linear system

$$\left(W + \lambda D^\top D \right) z = Wy, \quad (2)$$

where $W = \text{diag}(w)$ and D is the second-difference matrix. The weights are updated asymmetrically, $w_i = p$ if $y_i > z_i$ and $w_i = 1 - p$ otherwise, so that peaks above the baseline have little influence on it. We used $\lambda = 10^5$, $p = 0.01$ and 10 iterations, and subtracted z from y . Adaptive reweighted variants remove the need to tune p and are a natural next step.

3.3 Smoothing

High-frequency noise was reduced with a Savitzky–Golay filter [11], which fits a local polynomial of degree 3 in a sliding window of 11 points. This is equivalent to a convolution with fixed coefficients,

$$\hat{y}_i = \sum_{j=-5}^5 c_j y_{i+j}, \quad (3)$$

and preserves the height and width of narrow peaks better than a moving average.

3.4 Normalisation

Laser power, focus and local tissue density change the overall intensity of a spectrum. We removed this with standard normal variate (SNV) scaling [12],

$$\tilde{x}_i = \frac{x_i - \bar{x}}{s_x}, \quad \bar{x} = \frac{1}{L} \sum_i x_i, \quad s_x = \sqrt{\frac{1}{L} \sum_i (x_i - \bar{x})^2}. \quad (4)$$

SNV makes every spectrum zero-mean with unit variance. A side effect is that band intensities become relative: a strong increase in one band lowers all others after scaling. We return to this when interpreting the models.

3.5 Classifier

Pixels were classified with LightGBM [13], a gradient boosted decision tree method. For each class c the model builds an additive score

$$F_c(x) = \sum_{m=1}^M \eta f_{c,m}(x), \quad (5)$$

where each $f_{c,m}$ is a regression tree and η is the learning rate. Class probabilities come from the softmax

$$p(c | x) = \frac{\exp F_c(x)}{\sum_k \exp F_k(x)}, \quad (6)$$

and trees are fitted greedily to reduce the multiclass log loss $-\sum_i \log p(y_i | x_i)$. Gradient boosting handles non-linear interactions between bands and needs little tuning, which suited the time limit.

The base model used a learning rate of 0.05, at most 31 leaves and depth 6 per tree, 80% of features per tree and up to 300 trees with early stopping after 30 rounds without improvement. For the fingerprint window we also trained a more regularised variant: learning rate 0.01, depth 5, 70% feature and 80% row subsampling, L_1 and L_2 penalties of 0.5 and 1.0, balanced class weights and up to 1000 trees.

3.6 From pixels to samples

A diagnosis is made for a tissue sample, not for a pixel. For a map S with pixel spectra $x_1, \dots, x_N$ we averaged the predicted probabilities and took the most likely class:

$$\hat{P}(c | S) = \frac{1}{N} \sum_{i=1}^N p(c | x_i), \quad \hat{y}_S = \arg \max_c \hat{P}(c | S). \quad (7)$$

Averaging probabilities instead of counting hard votes lets confident pixels weigh more than uncertain ones.

3.7 Evaluation

Maps were split into training and test sets with `GroupShuffleSplit` using the map identifier as the group, so that pixels of one map never appeared on both sides. A fixed 20% of maps (24 per window) formed the test set. We report accuracy, balanced accuracy and macro-averaged F1. Because the test set is small, we give 95% Wilson score intervals [14] for accuracy,

$$\frac{\hat{p} + \frac{z^2}{2n}}{1 + \frac{z^2}{n}} \pm \frac{z}{1 + \frac{z^2}{n}} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n} + \frac{z^2}{4n^2}}, \quad z = 1.96, \quad (8)$$

where $\hat{p}$ is the observed accuracy and n is the number of test samples.

3.8 Spectral attribution

To find the bands the models rely on, we used the total split gain of each feature, that is the sum of loss reductions over all tree splits that use a given wavenumber, normalised to sum to one. We summed it over spectral ranges with known vibrational assignments [2, 15]. Gain is computed from the trained models alone and needs no data.

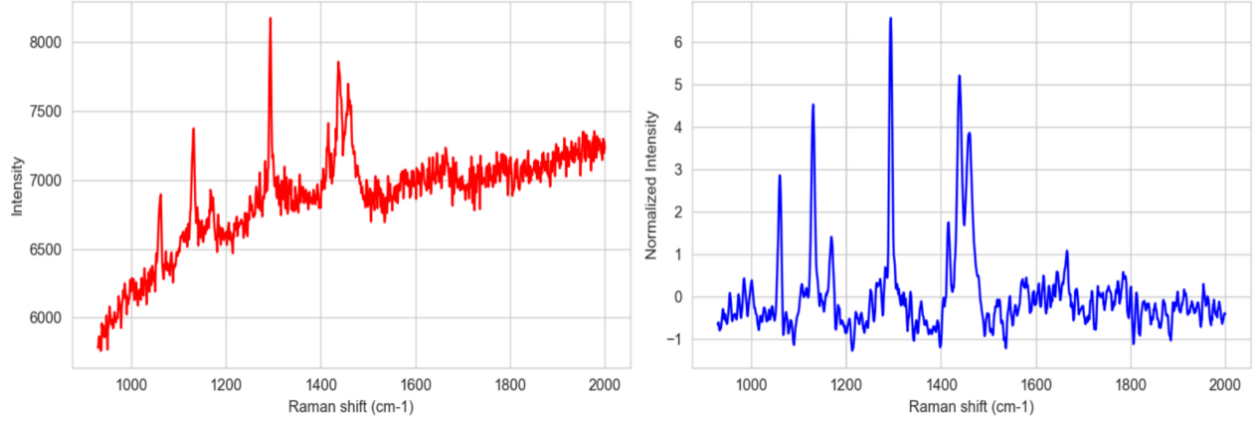


Figure 2: A single pixel spectrum from the fingerprint window before (left) and after (right) baseline removal, smoothing and SNV scaling. The sloping fluorescence background is removed and the Raman bands become comparable across pixels.

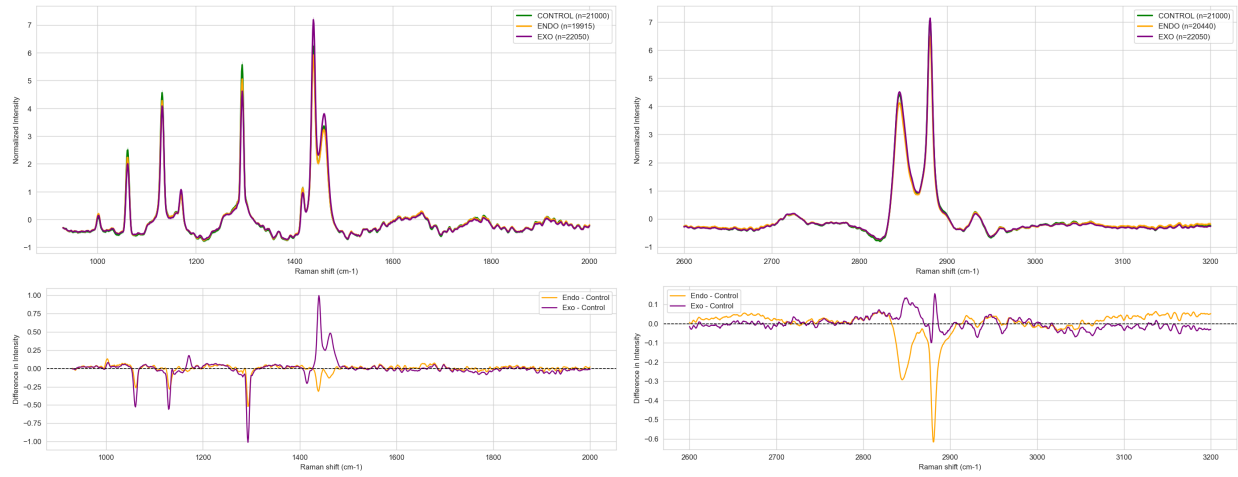


Figure 3: Top: mean preprocessed spectra of each group in the fingerprint (left) and high wavenumber (right) windows. Bottom: difference of the Endo and Exo means from the control mean.

4 Results

4.1 Preprocessing and spectral differences

Figure 2 shows the effect of preprocessing on one pixel. After cleaning, the group-average spectra (Figure 3, top) are very similar. Their differences from the control average (Figure 3, bottom) are small but structured and concentrated at a few bands. In the fingerprint window, Exo tissue shows a relative rise of the 1440 and 1460 cm^{-1} CH_2 deformation bands and a drop at 1062 , 1130 and 1296 cm^{-1} . Endo tissue shows smaller drops at 1062 , 1130 and 1296 cm^{-1} and, unlike Exo, also a drop at 1440 cm^{-1} . In the high wavenumber window the strongest effect is a loss of the CH_2 stretching bands at 2850 and 2880 cm^{-1} in Endo tissue, while Exo stays close to control, with a small rise near 2850 cm^{-1} and a derivative-shaped feature near 2880 cm^{-1} that suggests a slight band shift rather than a change in intensity.

Principal component analysis of the preprocessed spectra shows heavy overlap of the three groups in the first two components, which together explain only 29% of the variance (Figure 4).

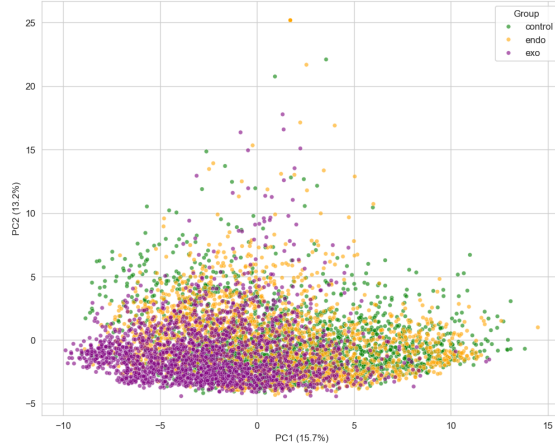


Figure 4: Projection of preprocessed fingerprint spectra onto the first two principal components (2000 random pixels per group).

Table 1: Classification results on the held-out maps. Pixel-level results are for individual spectra; sample-level results use the averaged map probability.

Model	Level	Accuracy (95% CI)	Balanced acc.	Macro F1
1500, base	pixel	0.61	—	0.60
1500, base	sample	0.83 (0.64–0.93)	0.84	0.83
1500, tuned	sample	0.75 (0.55–0.88)	—	0.74
2900, base	sample	0.79 (0.60–0.91)	0.79	0.78

The group signal is spread over many small, correlated changes rather than one dominant direction, which motivated a non-linear classifier.

4.2 Classification

Table 1 and Figure 5 summarise the results. At the level of single pixels the fingerprint model is right 61% of the time. Averaging over a map raises this to 83% for the fingerprint window and 79% for the high wavenumber window. The more regularised fingerprint model scored lower on the same split (75%). All intervals lie above chance, but they are wide: with 24 test maps, one extra error moves accuracy by four percentage points.

We show the pixel-level interval in Figure 5 to make a point: it is far too narrow. It treats the 13 090 test pixels as independent observations, while they come from only 24 maps and a handful of animals. This is a form of pseudoreplication [16], and it is the reason we base all conclusions on sample-level numbers.

The confusion matrices (Figure 6) show that the two windows make different kinds of errors. The fingerprint model recognised all six control maps but confused Endo and Exo with each other and with control. The high wavenumber model recognised all eight Exo maps but mixed up control and Endo.

4.3 Informative spectral regions

Figure 7 shows the gain attribution of the trained models across the spectrum, and Table 2 lists the share of gain for each band together with its usual assignment. Both models concentrate on a

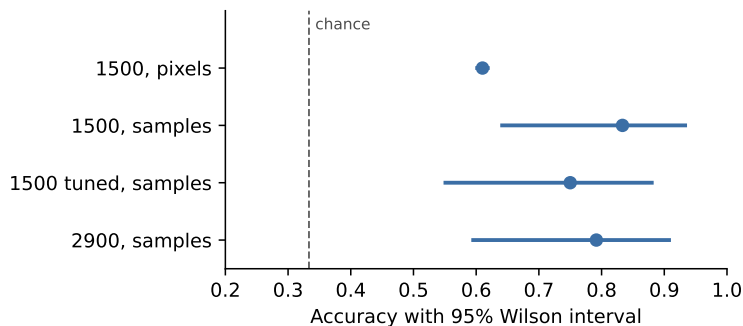


Figure 5: Accuracy with 95% Wilson intervals. The pixel-level interval assumes independent pixels and is far too narrow; see text.

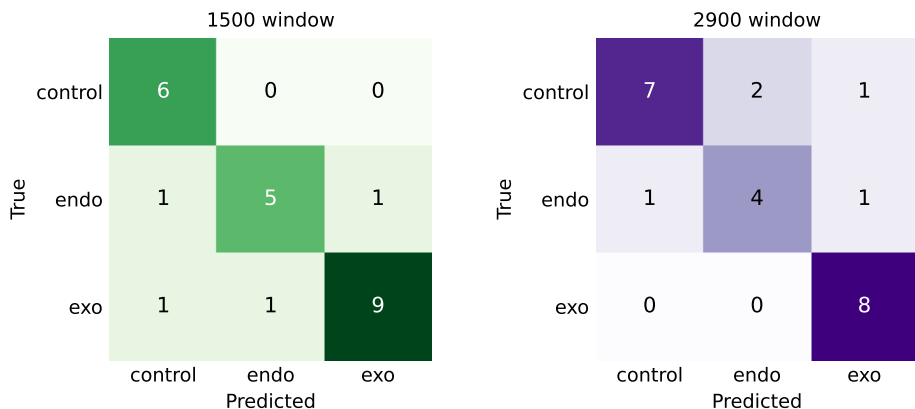


Figure 6: Sample-level confusion matrices on the held-out maps for the base models.

few narrow regions.

In the fingerprint window the largest share falls on the CH_2 scissoring band at about 1440 cm^{-1} (32%), followed by CH_2 twisting at about 1296 cm^{-1} (15%) and the C–C skeletal stretches of all-trans acyl chains at 1130 and 1062 cm^{-1} (12% together). These four bands are characteristic of lipids [15] and together receive almost 60% of the total gain. Bands that are specific to proteins, namely phenylalanine at 1003 cm^{-1} , amide III at $1230\text{--}1275\text{ cm}^{-1}$ and amide I at $1640\text{--}1680\text{ cm}^{-1}$, receive about 7%.

In the high wavenumber window the models use the asymmetric (2880 cm^{-1} , 33%) and symmetric (2850 cm^{-1} , 14%) CH_2 stretches of lipid chains, and the CH_3 stretch at about 2930 cm^{-1} (14%), which has a large protein contribution [2].

5 Discussion

5.1 A lipid signature, even in the fingerprint window

The fingerprint window is often described as the protein window, and our hackathon report followed that habit, assigning the band near 1290 cm^{-1} to amide III. The attribution analysis points the other way. In brain tissue, which is unusually rich in lipids, especially in myelin [17], the 1296 cm^{-1} region is dominated by CH_2 twisting of acyl chains, and the bands the model uses most are lipid

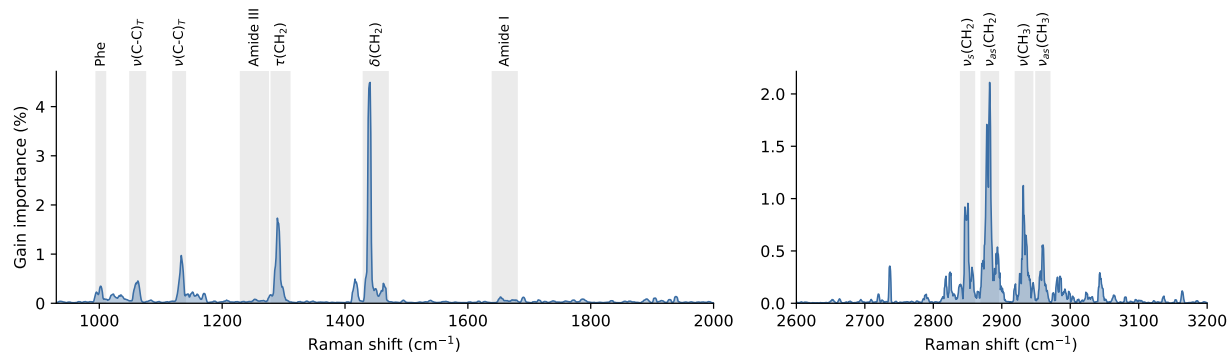


Figure 7: Split gain of the saved models across the fingerprint (left) and high wavenumber (right) windows, smoothed over five points. Shaded bands mark the assignments in Table 2. The fingerprint panel uses the regularised model shipped with the code.

Table 2: Share of total split gain per spectral band and the usual vibrational assignment [2, 15].

Window	Range (cm ⁻¹)	Assignment	Gain
1500	995–1010	Phenylalanine ring breathing (proteins)	2.9%
1500	1050–1075	C–C stretch, all-trans acyl chains (lipids)	4.6%
1500	1120–1140	C–C stretch, all-trans acyl chains (lipids)	7.2%
1500	1230–1275	Amide III (proteins)	1.9%
1500	1280–1310	CH ₂ twisting (mainly lipids)	14.8%
1500	1430–1470	CH ₂ scissoring (lipids, some protein)	32.0%
1500	1640–1680	Amide I (proteins)	2.1%
2900	2840–2860	Symmetric CH ₂ stretch (lipids)	13.6%
2900	2870–2895	Asymmetric CH ₂ stretch (lipids)	32.5%
2900	2920–2945	CH ₃ stretch (proteins and lipids)	14.3%
2900	2950–2970	Asymmetric CH ₃ stretch	6.6%

bands in both windows. Protein-specific bands carry little of the decision.

The direction of the changes is also informative. The 1062 and 1130 cm⁻¹ bands come from C–C stretching of chains in the all-trans conformation and are used as markers of lipid chain order [15]. Their relative decrease in Exo tissue, together with the rise of the CH₂ deformation bands, is consistent with a change in membrane packing. In the high wavenumber window, Endo tissue loses CH₂ stretching intensity, which may reflect lipid loss or remodelling under stress. Because SNV makes all intensities relative, these readings should be taken as hypotheses about ratios between bands, not as absolute concentrations.

These findings fit what is known about HSP70. Besides its chaperone role, HSP70 binds lipids and associates with membranes, where it can restore bilayer stability during stress [7]. A spectral signature that lives mostly in lipid bands is therefore plausible. We stress that this is a correlation found by a classifier and not a mechanistic result.

5.2 Two windows, two kinds of errors

The fingerprint model never missed a control map, and the high wavenumber model never missed an Exo map. This suggests a two-step scheme in which the fingerprint window first separates healthy

tissue from stressed tissue, and the high wavenumber window then decides between endogenous and exogenous HSP70. We could not test the combination, because the two windows were split independently and their test sets do not contain the same samples. A proper test needs paired maps of the same sample in both windows and a single split.

5.3 An audit of our own validation

Leakage between training and evaluation data is one of the most common reasons why machine learning results fail to reproduce [18], and spectroscopy is no exception [4]. Looking back at our protocol, we see three problems.

Splitting by map, not by animal. We kept pixels of one map together, which prevents the most obvious leak. But several maps come from the same animal (different regions and repeated scans), and there are only about ten animals in total. Maps of the same animal can therefore appear in both training and test sets, and the model can partly learn to recognise individual animals instead of experimental groups. For hierarchical data like this, the split should follow the highest level of the hierarchy that the model must generalise across, which here is the animal. Another team working on the same data used animal-level hold-outs, repeated over many animal triplets, and reported a macro F1 of about 0.41 and accuracy of about 0.46 for their final system [9]. The models and preprocessing differ, so the numbers are not directly comparable, but the gap is large enough to suggest that most of our margin over chance comes from animal-level leakage.

Early stopping and model choice on the test set. The held-out maps were passed to LightGBM as the early stopping set, and the base and regularised fingerprint models were compared on the same maps. Both steps let the test set influence the model, which biases the reported score upwards [19].

A single small split. With 24 test maps from a few animals, one random split gives a very noisy estimate [20]. The Wilson intervals in Table 1 already span more than 30 percentage points, and they still ignore the correlation between maps of the same animal.

Taken together, the accuracies in Table 1 are best read as optimistic upper bounds. The spectral attribution is less affected, since gain is a property of the fitted models and the same bands also stand out in the raw group differences of Figure 3, but it too would change under a stricter protocol.

5.4 Towards a leakage-free protocol

For this dataset we would now use the following setup:

1. leave-one-animal-out or repeated animal-level splits that hold out one animal per group, as in [9];
2. nested cross-validation, with early stopping and hyperparameter selection done only inside the training animals [4, 19];
3. a permutation test of the final score, in which group labels are shuffled at the animal level;
4. reporting per-region results, since brain regions differ strongly in lipid content and may confound group effects.

On the modelling side, natural extensions are SHAP values to explain individual decisions, interpretable ratio features such as I_{2880}/I_{2850} and I_{1130}/I_{1080} that track lipid chain packing and order [15], and one-dimensional convolutional networks.

6 Conclusion

A short pipeline of standard preprocessing steps and gradient boosting separates three HSP70-related groups of brain tissue on held-out Raman maps well above chance. The models rely mainly on lipid bands in both spectral windows, which suggests that membrane remodelling is the main spectral trace of the experimental conditions and fits the known membrane biology of HSP70. At the same time, our evaluation lets maps of the same animal appear in both training and test data and reuses the test set for model choice, so the headline accuracies are upper bounds. The most useful next step is not a stronger model but an animal-level, nested evaluation of the one we have.

Code and data availability

Code and trained models are available at https://github.com/igor688-hub/Nuclear-IT_Hack. The Raman data were provided by the hackathon organisers and are not publicly available.

Acknowledgements

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